

CUSTOMER RETENTION INTELLIGENCE SYSTEM

Churn AI Web Application

System Walkthrough & Screen Documentation

A guided tour of each screen in the deployed Churn AI dashboard, describing its purpose, layout, and the functionality it provides to telecom operations users.

1. Introduction

This document presents the user-facing web application built for the Customer Retention Intelligence System project. The application, branded “Churn AI — Telecom Analytics,” gives telecom operations staff a single workspace for reviewing churn analytics, searching individual customers, identifying high-risk accounts, and generating an on-demand churn prediction for a given customer profile.

The sections below walk through each screen of the application in the order a user would typically encounter them — starting from sign-in, through the analytics dashboard, and into the customer-level and prediction tools. Each screen is shown as a screenshot followed by a description of its layout and functionality.

Authentication / Sign-In Page

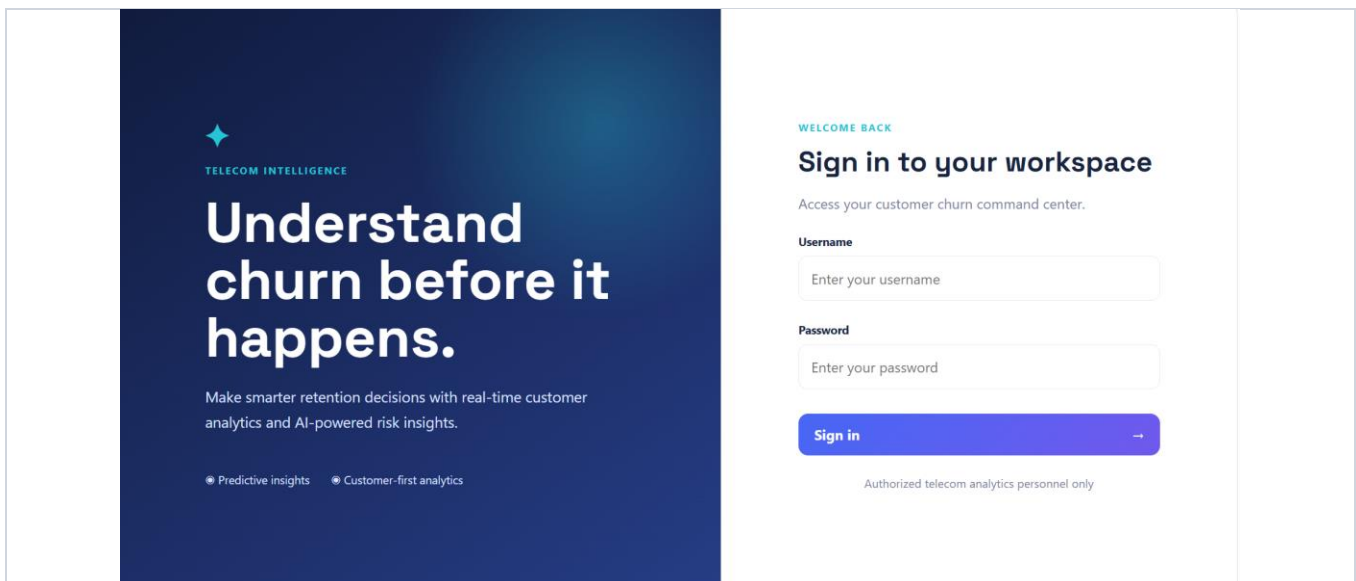


Figure 1. The Churn AI sign-in screen.

This is the entry point of the application. The page is split into two halves: a branded left panel that introduces the product (“Understand churn before it happens”) and communicates its value proposition — real-time customer analytics and AI-powered risk insights — and a right-hand authentication panel where the user signs in.

Functionality

- Username and password fields, each with placeholder guidance text.
- A primary “Sign in” call-to-action button that authenticates the user against the backend and, on success, redirects into the workspace.
- A footer notice (“Authorized telecom analytics personnel only”) indicating the tool is restricted to internal staff, i.e. access control is enforced at login.
- Two feature badges (“Predictive insights” and “Customer-first analytics”) that reinforce the tool's purpose before the user even signs in.

Churn Analytics Dashboard (Overview)

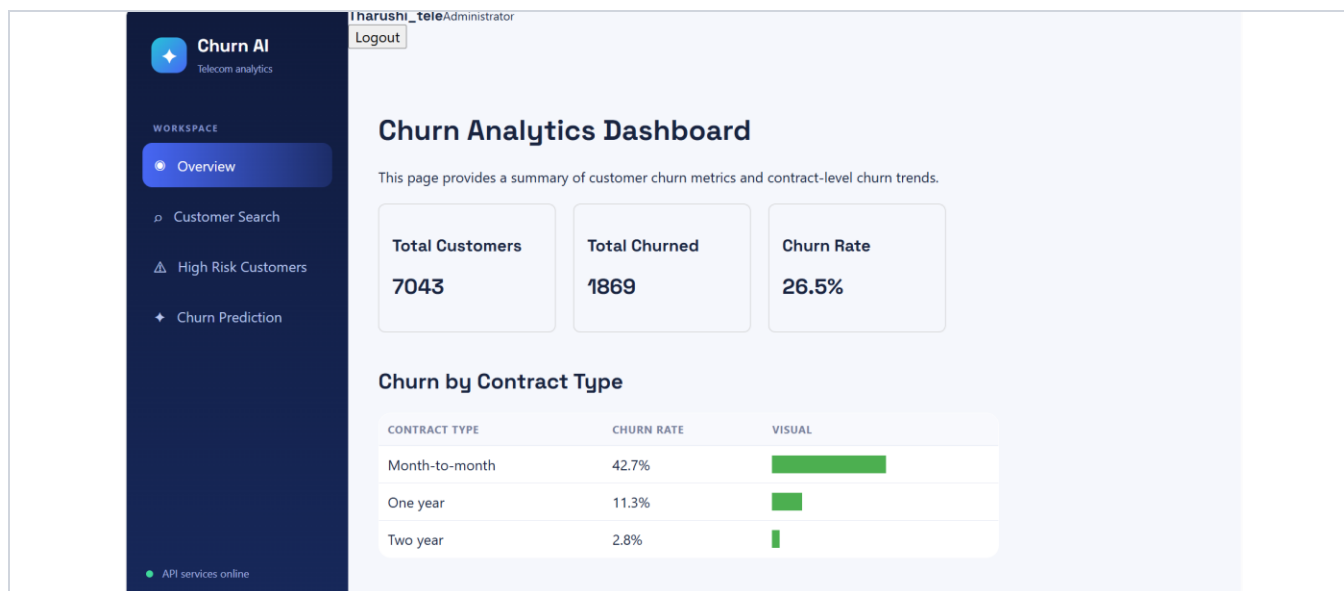


Figure 2. The Overview page — the default landing screen after login.

Once authenticated, the user lands on the Overview page, which is the default selection in the left-hand “WORKSPACE” navigation menu. This page summarizes the entire customer base at a glance and highlights how churn varies by contract type.

Functionality

- A persistent left sidebar (present on every page) showing the product name, a “WORKSPACE” navigation section with links to Overview, Customer Search, High Risk Customers and Churn Prediction, and a live “API services online” status indicator at the bottom.
- A top bar showing the signed-in user (“Tharushi_tele”), their role (“Administrator”), a notification bell, and a Logout control.
- Three summary KPI cards: Total Customers (7,043), Total Churned (1,869), and the overall Churn Rate (26.5%) — giving an instant, top-level health check of retention.
- A “Churn by Contract Type” table that breaks the churn rate down by contract (Month-to-month: 42.7%, One year: 11.3%, Two year: 2.8%), each with a proportional colored bar for quick visual comparison.
- This breakdown surfaces the key business insight that customers on flexible, month-to-month contracts churn at a far higher rate than those locked into annual or two-year agreements — directly informing retention strategy.

Customer Search

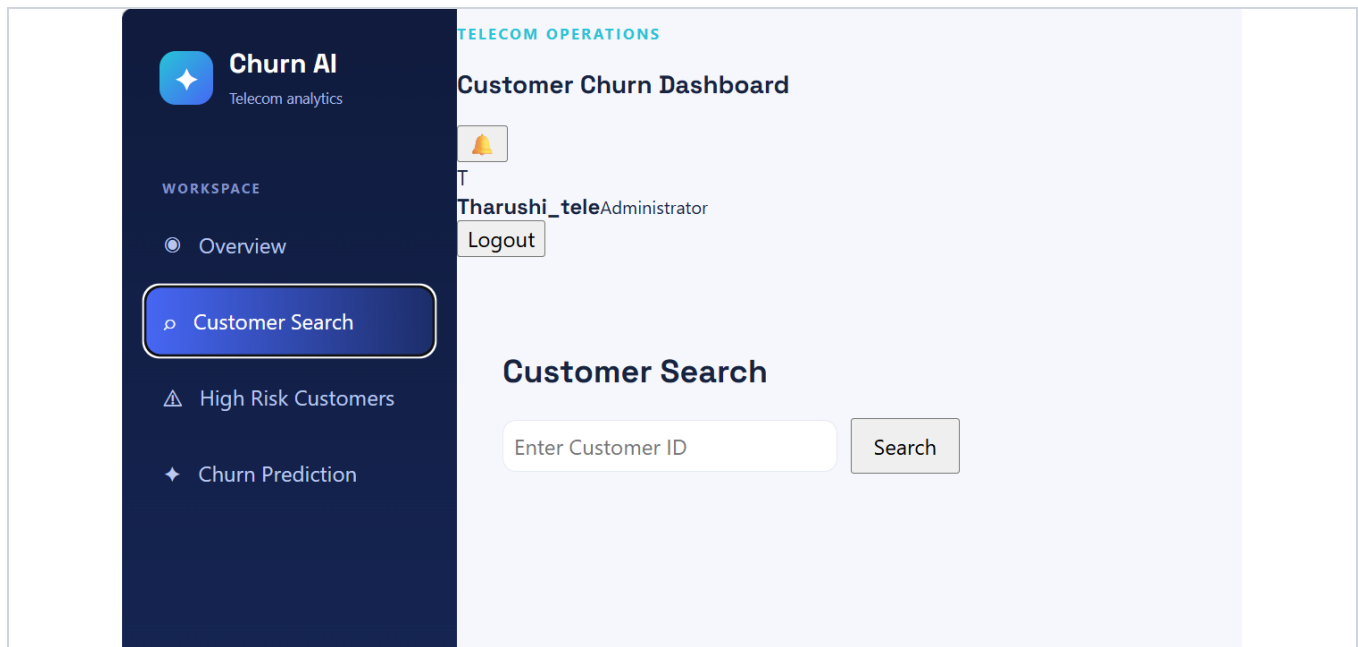


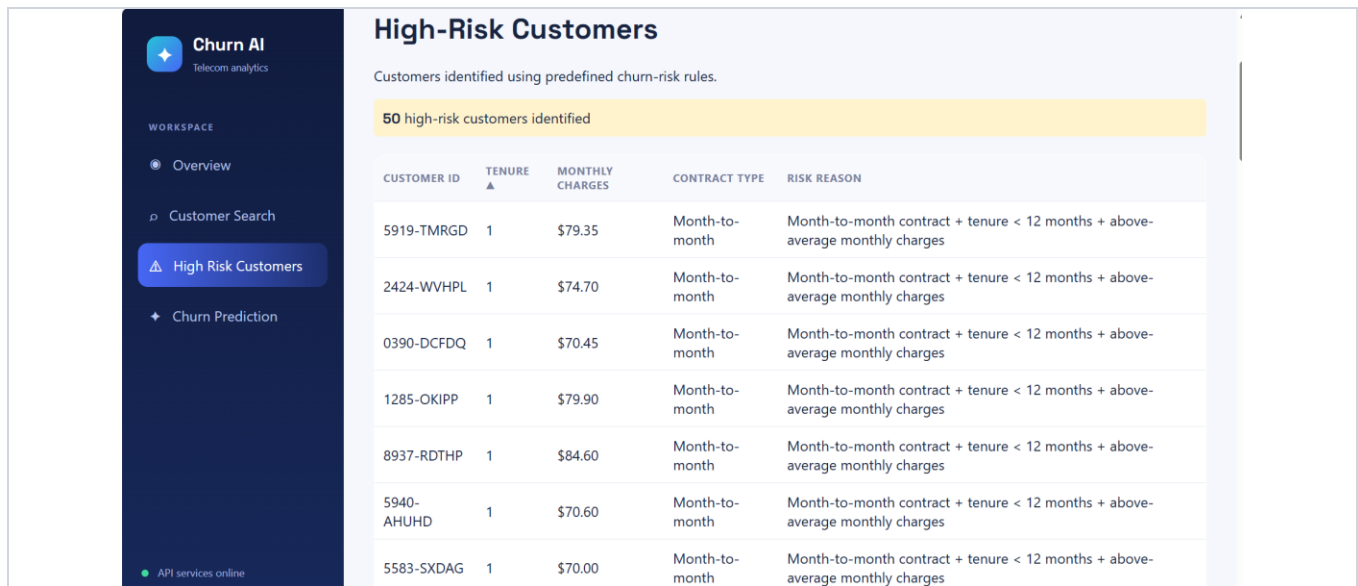
Figure 3. The Customer Search screen.

Selected from the sidebar, this screen lets a user look up an individual customer record by their unique identifier rather than browsing the full dataset.

Functionality

- A single text input (“Enter Customer ID”) for typing or pasting a customer's ID.
- A “Search” button that queries the backend for that specific customer and would display their profile and churn-related attributes (tenure, charges, contract, services, current risk status, etc.) once a valid ID is submitted.
- This screen is the primary tool for support and retention staff who already know which customer they need to investigate, complementing the aggregate view on the Overview page and the pre-filtered list on High Risk Customers.

High-Risk Customers



CUSTOMER ID	TENURE ▲	MONTHLY CHARGES	CONTRACT TYPE	RISK REASON
5919-TMRGD	1	\$79.35	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
2424-WVHPL	1	\$74.70	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
0390-DCFDQ	1	\$70.45	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
1285-OKIPP	1	\$79.90	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
8937-RDTHP	1	\$84.60	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
5940-AHUHD	1	\$70.60	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges
5583-SXDAG	1	\$70.00	Month-to-month	Month-to-month contract + tenure < 12 months + above-average monthly charges

Figure 4. The High-Risk Customers list, filtered by predefined churn-risk rules.

This page proactively surfaces the customers most likely to churn, rather than requiring a user to search for them individually. A banner at the top states how many customers currently meet the risk criteria (50 high-risk customers identified).

Functionality

- A sortable table listing each flagged customer's ID, tenure (in months), monthly charges, contract type, and the specific reason they were flagged.
- Risk is determined by a transparent, rule-based combination of factors visible in the “Risk Reason” column — in the examples shown, customers are flagged when they are on a month-to-month contract, have less than 12 months of tenure, and pay above-average monthly charges.
- The Tenure column header shows a sort indicator (▲), meaning the list can be reordered — here it is sorted ascending, surfacing the newest, most at-risk customers first.
- Because the flagging logic is rule-based and explainable, this page gives retention teams a ready-made, prioritized call list without needing to interpret a machine-learning model directly.

Churn Prediction Tool

The screenshot displays the 'Churn Prediction' interface. On the left, a dark blue sidebar contains the 'Churn AI' logo and a 'Workspace' menu with options: Overview, Customer Search, High Risk Customers, and Churn Prediction (which is highlighted with a blue border and a plus icon). At the bottom of the sidebar, a green dot indicates 'API services online'. The top of the main content area shows the user 'Tharushi_teleAdministrator' and a 'Logout' button. The main heading is 'Churn Prediction', followed by the instruction 'Enter customer details to generate a churn risk prediction.' The form includes four input fields: 'Tenure' (text), 'Monthly Charges' (text), 'Contract Type' (a dropdown menu currently showing 'Month-to-month'), and 'Service Count' (text). A 'Predict Churn' button is located at the bottom of the form.

Figure 5. The Churn Prediction form, used to score a customer profile on demand.

This is the AI-powered feature of the application. Rather than relying only on static rules, it lets a user enter a customer profile and get a churn-risk prediction back from the underlying machine-learning model in real time.

Functionality

- An input form collecting the key predictive features: Tenure (months with the company), Monthly Charges, Contract Type (a dropdown — Month-to-month, One year, Two year), and Service Count (number of services subscribed to).
- A “Predict Churn” button that submits these values to the prediction API and returns a churn-risk result for that hypothetical or real customer.
- This tool is useful for what-if analysis — for example, testing how a proposed contract change or service add-on might shift a specific customer's churn risk — as well as for scoring new customers who are not yet in the flagged high-risk list.

2. Summary

Together, these five screens form a complete retention workflow: Overview gives leadership a real-time health check of churn across the customer base; Customer Search and High-Risk Customers let operations staff drill into specific accounts, either by lookup or by proactive, rule-based flagging; and Churn Prediction adds a forward-looking, AI-driven layer that scores any customer profile on demand. The consistent sidebar navigation, live API status indicator, and role-aware top bar (Administrator access, Logout) tie the experience together as a single, secured analytics workspace for telecom churn management.